

DBMS Lab 1 - Books Database

DBMS Lab 1 — Books Database (Work-wise Breakdown)

Objective: Design and implement a relational database `books` consisting of three related tables — `Author`, `Books`, and `Publication` — apply primary/foreign key constraints, insert sample records, and perform various `SELECT`, `UPDATE`, and `DELETE` operations.

Each piece of work below is shown as: the **original task/question**, followed by the **SQL terminal** (input + output) that solves it.

Work 0 — Schema Given & Modifications

Given schema: - `Books (b_title, b_gener, b_price, b_ISBN, b_author)` - `Author(a_name, a_address, a_contact, a_numberOfBooks, a_id)` - `Publication(p_id, p_name, p_address, p_established_date)`

Change	Reason
<code>b_ISBN</code> made Primary Key of <code>Books</code>	Unique book identifier
<code>a_id</code> made Primary Key of <code>Author</code>	Needed as FK reference target
<code>b_author</code> in <code>Books</code> set as Foreign Key → <code>Author(a_id)</code>	Task requires “add foreign key author constraint”
Added <code>a_gender</code> column to <code>Author</code>	Queries ask for “lady authors” / “male authors from Kathmandu”
<code>p_id</code> made Primary Key of <code>Publication</code>	Needed for later <code>UPDATE ... WHERE p_id</code>
<code>b_gener</code> → <code>b_genre</code> , <code>p_established_date</code> → <code>p_establishedDate</code>	Spelling cleanup

Work 1 — Create Database and Tables

Task: Create the database and the three tables (`Author`, `Books`, `Publication`) with appropriate primary keys, foreign key constraint, and check constraint.

```
mysql> CREATE DATABASE book;
Query OK, 1 row affected (0.05 sec)

mysql> USE book;
Database changed

mysql> CREATE TABLE Author (
->   a_id INT PRIMARY KEY AUTO_INCREMENT,
->   a_name VARCHAR(50) NOT NULL,
->   a_address VARCHAR(50),
->   a_contact VARCHAR(15),
->   a_numberOfBooks INT DEFAULT 0,
->   a_gender ENUM('Male','Female') NOT NULL
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Books (
->   b_ISBN VARCHAR(20) PRIMARY KEY,
->   b_title VARCHAR(100) NOT NULL,
->   b_genre VARCHAR(30),
->   b_price DECIMAL(8,2) CHECK (b_price > 0),
->   b_author INT,
->   CONSTRAINT fk_author
->     FOREIGN KEY (b_author)
->     REFERENCES Author(a_id)
->     ON DELETE CASCADE
->     ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Publication (
->   p_id INT PRIMARY KEY AUTO_INCREMENT,
->   p_name VARCHAR(50) NOT NULL,
->   p_address VARCHAR(50),
->   p_establishedDate DATE
```

```

-> );
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_book |
+-----+
| Author          |
| Books           |
| Publication      |
+-----+
3 rows in set (0.01 sec)

```

Work 2 — Insert Records (5+ into each table)

Task: Insert at least 5 sample records into each of the three tables.

```

mysql> INSERT INTO Author
-> (a_name, a_address, a_contact, a_numberOfBooks, a_gender)
-> VALUES
-> ('Sita Sharma', 'Kathmandu', '9841000111', 3, 'Female'),
-> ('Ram Thapa', 'Kathmandu', '9841000222', 2, 'Male'),
-> ('Radha Gurung', 'Pokhara', '9841000333', 4, 'Female'),
-> ('Rajesh KC', 'Lalitpur', '9841000444', 1, 'Male'),
-> ('Bibek Koirala', 'Kathmandu', '9841000555', 2, 'Male'),
-> ('Anita Rai', 'Biratnagar', '9841000666', 1, 'Female');
Query OK, 6 rows affected (0.02 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Author;
+-----+-----+-----+-----+-----+-----+
| a_id | a_name      | a_address | a_contact | a_numberOfBooks | a_gender |
+-----+-----+-----+-----+-----+-----+
| 1 | Sita Sharma | Kathmandu | 9841000111 | 3 | Female |
| 2 | Ram Thapa   | Kathmandu | 9841000222 | 2 | Male   |
| 3 | Radha Gurung | Pokhara   | 9841000333 | 4 | Female |
| 4 | Rajesh KC   | Lalitpur  | 9841000444 | 1 | Male   |
| 5 | Bibek Koirala | Kathmandu | 9841000555 | 2 | Male   |
| 6 | Anita Rai   | Biratnagar | 9841000666 | 1 | Female |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> INSERT INTO Books
-> (b_ISBN, b_title, b_genre, b_price, b_author)
-> VALUES
-> ('ISBN001', 'Nepali Kavita Sangraha', 'Poetry', 450.00, 1),
-> ('ISBN002', 'Modern History Nepal', 'History', 750.00, 2),
-> ('ISBN003', 'Pahilo Prem', 'Novel', 620.00, 3),
-> ('ISBN004', 'Computer Networks Basics', 'Technology', 850.00, 4),
-> ('ISBN005', 'Kathmandu Katha', 'Fiction', 300.00, 5),
-> ('ISBN006', 'Yatra Sansmaran', 'Travel', 700.00, 6);
Query OK, 6 rows affected (0.05 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+-----+
| b_ISBN | b_title                | b_genre | b_price | b_author |
+-----+-----+-----+-----+-----+
| ISBN001 | Nepali Kavita Sangraha | Poetry  | 450.00 | 1 |
| ISBN002 | Modern History Nepal   | History | 750.00 | 2 |
| ISBN003 | Pahilo Prem            | Novel   | 620.00 | 3 |
| ISBN004 | Computer Networks Basics | Technology | 850.00 | 4 |
| ISBN005 | Kathmandu Katha        | Fiction | 300.00 | 5 |
| ISBN006 | Yatra Sansmaran        | Travel  | 700.00 | 6 |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> INSERT INTO Publication
-> (p_name, p_address, p_establishedDate)
-> VALUES
-> ('Sajha Prakashan', 'Kathmandu', '1965-01-01'),
-> ('Ratna Pustak Bhandar', 'Kathmandu', '1950-05-10'),
-> ('Fine Print', 'Kathmandu', '2010-03-15'),
-> ('Vidyarthi Pustak Bhandar', 'Bhaktapur', '2005-07-20'),
-> ('Nepa Publications', 'Lalitpur', '2015-09-01');
Query OK, 5 rows affected (0.05 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Publication;
+-----+-----+-----+

```

```
| p_id | p_name | p_address | p_establishedDate |
+-----+-----+-----+-----+
| 1 | Sajha Prakashan | Kathmandu | 1965-01-01 |
| 2 | Ratna Pustak Bhandar | Kathmandu | 1950-05-10 |
| 3 | Fine Print | Kathmandu | 2010-03-15 |
| 4 | Vidyarthi Pustak Bhandar | Bhaktapur | 2005-07-20 |
| 5 | Nepa Publications | Lalitpur | 2015-09-01 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Work 3 — Query: Books priced above 600

Task: Display all books where the price is more than 600.

```
mysql> SELECT * FROM Books
-> WHERE b_price > 600;
+-----+-----+-----+-----+-----+
| b_ISBN | b_title | b_genre | b_price | b_author |
+-----+-----+-----+-----+-----+
| ISBN002 | Modern History Nepal | History | 750.00 | 2 |
| ISBN003 | Pahilo Prem | Novel | 620.00 | 3 |
| ISBN004 | Computer Networks Basics | Technology | 850.00 | 4 |
| ISBN006 | Yatra Sansmaran | Travel | 700.00 | 6 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Work 4 — Query: List all lady (female) authors

Task: List all details of lady (female) authors only.

```
mysql> SELECT * FROM Author
-> WHERE a_gender = 'Female';
+-----+-----+-----+-----+-----+-----+
| a_id | a_name | a_address | a_contact | a_numberOfBooks | a_gender |
+-----+-----+-----+-----+-----+-----+
| 1 | Sita Sharma | Kathmandu | 9841000111 | 3 | Female |
| 3 | Radha Gurung | Pokhara | 9841000333 | 4 | Female |
| 6 | Anita Rai | Biratnagar | 9841000666 | 1 | Female |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Work 5 — Query: Male authors from Kathmandu

Task: List all male authors from Kathmandu.

```
mysql> SELECT * FROM Author
-> WHERE a_gender = 'Male'
-> AND a_address = 'Kathmandu';
+-----+-----+-----+-----+-----+-----+
| a_id | a_name | a_address | a_contact | a_numberOfBooks | a_gender |
+-----+-----+-----+-----+-----+-----+
| 2 | Ram Thapa | Kathmandu | 9841000222 | 2 | Male |
| 5 | Bibek Koirala | Kathmandu | 9841000555 | 2 | Male |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Work 6 — Query: Name starts with 'R', address not Kathmandu

Task: Display all records where the author's name starts with 'R' and address is not Kathmandu.

```
mysql> SELECT * FROM Author
-> WHERE a_name LIKE 'R%'
-> AND a_address != 'Kathmandu';
+-----+-----+-----+-----+-----+-----+
| a_id | a_name      | a_address | a_contact | a_numberOfBooks | a_gender |
+-----+-----+-----+-----+-----+-----+
| 3 | Radha Gurung | Pokhara   | 9841000333 | 4 | Female |
| 4 | Rajesh KC    | Lalitpur  | 9841000444 | 1 | Male   |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Work 7 — Query: Publications ordered by established date

Task: Display all records of publication in order (by established date).

```
mysql> SELECT * FROM Publication
-> ORDER BY p_establishedDate;
+-----+-----+-----+-----+-----+
| p_id | p_name      | p_address | p_establishedDate |
+-----+-----+-----+-----+-----+
| 2 | Ratna Pustak Bhandar | Kathmandu | 1950-05-10 |
| 1 | Sajha Prakashan      | Kathmandu | 1965-01-01 |
| 4 | Vidhyarthi Pustak Bhandar | Bhaktapur | 2005-07-20 |
| 3 | Fine Print           | Kathmandu | 2010-03-15 |
| 5 | Nepa Publications    | Lalitpur  | 2015-09-01 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Work 8 — Query: Only id and established date

Task: Display id and established date only from the Publication table.

```
mysql> SELECT p_id, p_establishedDate
-> FROM Publication;
+-----+-----+
| p_id | p_establishedDate |
+-----+-----+
| 1 | 1965-01-01 |
| 2 | 1950-05-10 |
| 3 | 2010-03-15 |
| 4 | 2005-07-20 |
| 5 | 2015-09-01 |
+-----+-----+
5 rows in set (0.00 sec)
```

Work 9 — Update: Change established dates for p_id 2 and 3

Task: Change the established date of publications where p_id is 2 and 3.

```
mysql> UPDATE Publication
-> SET p_establishedDate = '2012-06-15'
-> WHERE p_id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Publication
-> SET p_establishedDate = '2011-01-01'
-> WHERE p_id = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM Publication;
+-----+-----+-----+-----+
| p_id | p_name      | p_address | p_establishedDate |
+-----+-----+-----+-----+
| 1 | Sajha Prakashan      | Kathmandu | 1965-01-01 |
| 2 | Ratna Pustak Bhandar | Kathmandu | 2012-06-15 |
| 3 | Vidhyarthi Pustak Bhandar | Bhaktapur | 2011-01-01 |
| 4 | Fine Print           | Kathmandu | 2010-03-15 |
| 5 | Nepa Publications    | Lalitpur  | 2015-09-01 |
+-----+-----+-----+-----+
```

```
+-----+-----+-----+-----+
| 1 | Sajha Prakashan      | Kathmandu | 1965-01-01 |
| 2 | Ratna Pustak Bhandar  | Kathmandu | 2012-06-15 |
| 3 | Fine Print            | Kathmandu | 2011-01-01 |
| 4 | Vidyarthi Pustak Bhandar | Bhaktapur | 2005-07-20 |
| 5 | Nepa Publications     | Lalitpur  | 2015-09-01 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Work 10 — Delete: Remove publications established before 2009

Task: Delete all records from Publication whose established date is before 2009.

```
mysql> DELETE FROM Publication
-> WHERE p_establishedDate < '2009-01-01';
Query OK, 2 rows affected (0.01 sec)

mysql> SELECT * FROM Publication;
+-----+-----+-----+-----+
| p_id | p_name          | p_address | p_establishedDate |
+-----+-----+-----+-----+
| 2 | Ratna Pustak Bhandar | Kathmandu | 2012-06-15 |
| 3 | Fine Print          | Kathmandu | 2011-01-01 |
| 5 | Nepa Publications   | Lalitpur  | 2015-09-01 |
+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

Note: *p_id = 2* was originally 1950-05-10 (before 2009), but it was updated to 2012-06-15 in **Work 9** before this delete ran — so it survives. This is a good demonstration of how an *UPDATE* earlier in the sequence can change which rows a later *DELETE ... WHERE* condition matches.

Quick Revision Summary

Work #	Task	SQL Clause Used
1	Create DB & 3 tables (PK, FK, CHECK)	CREATE DATABASE, CREATE TABLE
2	Insert ≥5 records into each table	INSERT INTO ... VALUES
3	Books priced > 600	SELECT ... WHERE b_price > 600
4	List female authors	WHERE a_gender = 'Female'
5	Male authors from Kathmandu	WHERE ... AND ...
6	Name starts with 'R', not Kathmandu	LIKE 'R%' AND !=
7	Publications sorted by date	ORDER BY p_establishedDate
8	Only id + established date	SELECT p_id, p_establishedDate
9	Update dates for p_id 2, 3	UPDATE ... SET ... WHERE
10	Delete pre-2009 publications	DELETE FROM ... WHERE